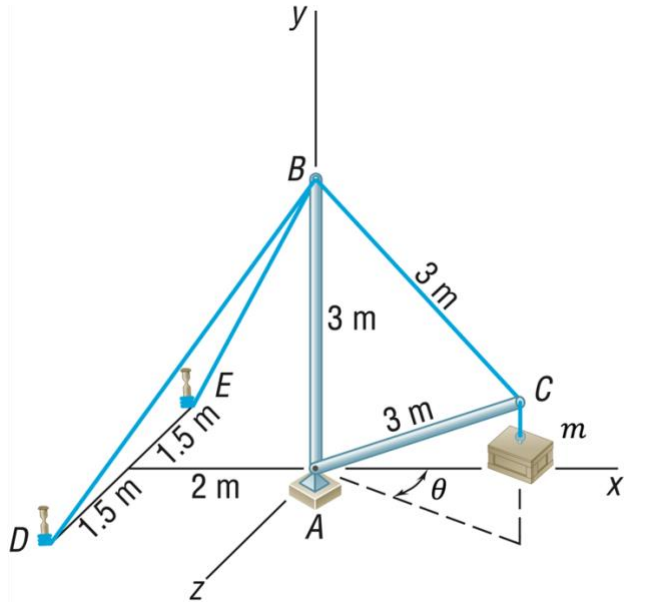


Quiz 02. Force systems and static equilibrium in 3D space
prepared by C. Cahig for ENS161 (AY 2026-2027, S1)

Instructions. Unless otherwise stated, all quantities are in SI units. Take the acceleration due to gravity to be 9.80665.

An m -tonne load is supported by a derrick. Held in place by cables connected to the ground at D and E , the derrick is designed so that it approximately lies in a vertical plane that forms an angle θ with the xy -plane. Model the base of the derrick is as a ball-and-socket joint. Forces are to be in kN and moments in kN-m.



Scenario 1. For $m = 1$ and $\theta = \frac{\pi}{6}$, solve for the following moments (as Cartesian vectors) due to the load.

1. about A
2. at B
3. about E
4. at D
5. about the axis through A and D
6. about the line through A and E
7. about the axis through D and E

Scenario 2. For $m = 1$ and $\theta = \frac{\pi}{3}$, solve for:

1. the tension in cable BD ,
2. the tension in cable BE ,
3. the magnitude of the reaction R_A exerted by the ball-and-socket joint onto the derrick,
4. the tension in cable BC ,
5. the force developed in member AC , and
6. the force developed in member AB .
7. Is member AC in compression?
8. Is member AB in compression?

Scenario 3. When $\theta = \frac{\pi}{4}$, solve for:

1. the tension in cable BD ,
 2. the tension in cable BE ,
 3. the magnitude of R_A ,
 4. the tension in cable BC ,
 5. the force developed in member AC , and
 6. the force developed in member AB
- per tonne of load.

Marking. Each numbered item above is worth three (3) points. The total percentage rating a student gets for this Activity is computed as $0.59(0.9^m r) + 0.41p$, where m is the number of unmet requirements and violated instructions, r is the raw score (in percent), and p is the peer rating (in percent) by his/her groupmates.

Submission. Answers shall be communicated through the accompanying Google Form. Peer ratings shall be communicated via the submission platform in the Google Classroom.

Thoughts to be think

- For a given load, what values of θ will minimize and maximize the tension in cable BD ? in cable BE ? in cable BC ?
- If strength limits (e.g., maximum tensions the cables could safely sustain) are given, what would be the maximum allowable load for a given θ ?
- If strength limits (e.g., maximum tensions the cables could safely sustain) are given, what would be the allowable values of θ for a given load?